

Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech IV Semester - End Semester Examination - April / May -2024

### Python Programming [CSE-4309]

**Marks: 50**

**Duration: 180 mins.**

#### PART-A

**Answer all the questions.**

- 1) Discuss the importance of comments, variables, and assignments in Python programming. (4)
  - A)
  - B) Write a Python program to determine if a given year is a leap year or not. Prompt the user to enter a year, and then check if it's a leap year. Print "Leap year" if it is, and "Not a leap year" otherwise. Remember, a leap year is either divisible by 4 but not divisible by 100, or divisible by 400. (4)
  - C) Write a Python program to create a dictionary where the keys are numbers from 1 to 5, and the values are the squares of those numbers, using dictionary comprehension. (2)
- 2) Explain the concept of packing and unpacking tuples in Python. Provide examples illustrating both packing and unpacking operations. (4)
  - A)
  - B) Explain the concept of sets and their significance in mathematics. Provide examples to illustrate the basic properties of sets such as union, intersection, and complement. How are sets different from other data structures like lists or arrays? (4)
  - C) Distinguish between the split() and join() methods in Python strings. (2)
- 3) Create a simple Python module consisting of at least two functions and demonstrate how to import and use this module in another Python script. (4)
  - A)
  - B) Discuss the importance of widget placement and layout management in Tkinter applications, highlighting different geometry managers and their respective advantages. (4)
  - C) Explain the concept of class variables in Python and demonstrate their usage with example. (2)

- 4) Describe how array indexing works in NumPy. Provide an example illustrating slicing and indexing in a 1D array. (4)
- A)
- B) Write a Python function that reads a text file and returns the number of words in it. (4)
- C) Explain the purpose of a function for reading text data in a programming language. Describe one scenario where such a function would be useful. (2)
- 5) Discuss the origin of the term "dictionary" in relation to programming and elucidate its relevance in data structures. (4)
- A)
- B) Describe a scenario where counting words in Python could be useful in real-world applications, and explain how you would implement it. (2)
- C) What are the key features of Google App Engine (GAE), and how does it differ from traditional web hosting solutions? (4)

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